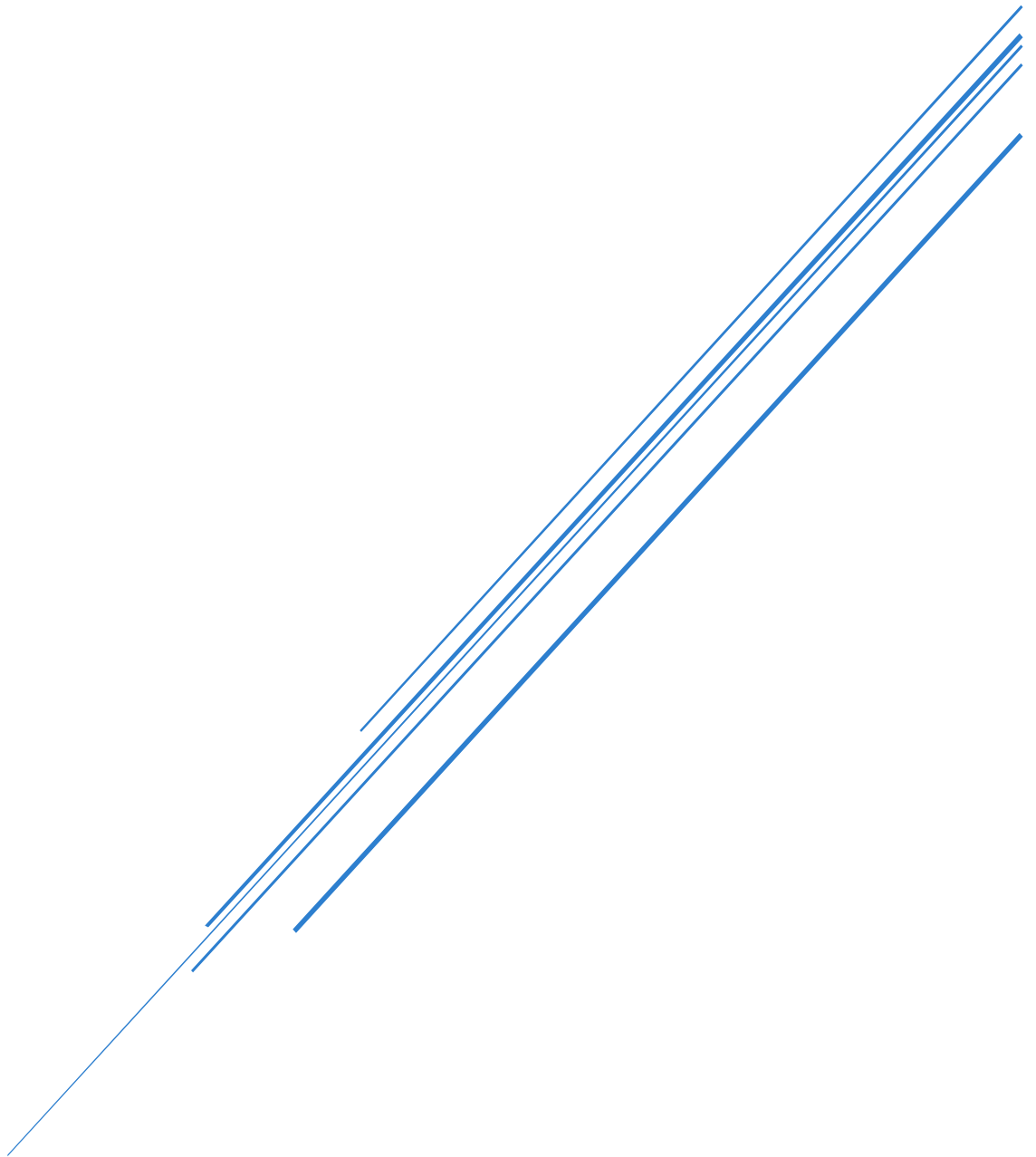


SYNAPSE STOCKS

AI-POWERED STOCK INTELLIGENCE PLATFORM



PAWGRAMMERS TEAM
Vibe Coding Project

SYNAPSE STOCKS: AI-POWERED STOCK INTELLIGENCE PLATFORM

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SYNAPSE STOCKS: AI-POWERED STOCK INTELLIGENCE PLATFORM

Welcome to SynapseStocks

A full-stack, agentic AI platform built to help investors and analysts decode the stock market with clarity and confidence.

Whether you're tracking financial trends, climate impact, or social sentiment, SynapseStocks empowers you with explainable, multi-factor insights through:

- Modular AI agents tailored to different sectors
- A sleek, responsive dashboard built with React and Material-UI
- Secure JWT-based authentication
- A clean monorepo structure for easy development and deployment

This project is ideal for developers, data scientists, and financial analysts looking to explore intelligent market analysis tools in a demo or educational setting.

SYNAPSE STOCKS: AI-POWERED STOCK INTELLIGENCE PLATFORM

Impact

SynapseStocks isn't just another stock analysis tool—it's a smart companion for investors who want to make sense of the market's complexity. Whether you're just starting out or you're a seasoned analyst, the platform helps you turn raw data into meaningful insights with the help of intelligent, modular AI agents and a sleek, intuitive dashboard.

What Makes It Stand Out

- **Smarter Decisions, Made Simple**
SynapseStocks delivers clear, multi-factor insights that go beyond basic charts and numbers—so users can make confident, well-informed investment choices.
- **Insights That Speak Your Language**
From climate trends to chemical pricing and regulatory shifts, each AI agent is tuned to specific sectors, helping users understand how real-world events impact stock performance.
- **Built for Everyone**
With a lightweight backend and clean UI, it's perfect for students, educators, and developers looking to explore financial data or build prototypes.
- **Ready to Grow**
Thanks to its modular design, new agents and data sources can be added easily—making it a future-proof solution for evolving market needs.

Implementation

SynapseStocks is crafted to be developer-friendly and easy to scale. Its monorepo setup keeps everything organized, while the clear separation between frontend and backend makes updates and maintenance a breeze.

Under the Hood

- **Frontend**
Built with React and Material-UI, the dashboard is fast, responsive, and designed to make complex data feel approachable.

PROJECT IMPACT & IMPLEMENTATION

SYNAPSE STOCKS: AI-POWERED STOCK INTELLIGENCE PLATFORM

- **Backend**
The Node.js + Express server handles everything from API calls to agent logic and user authentication. Data is stored using lowdb for simplicity.
- **Secure Login**
JWT-based authentication ensures that users can safely access personalized insights and agent configurations.
- **Plug-and-Play AI Agents**
Each agent is modular and easy to extend. Want to add a new market factor? Just plug in a new agent—no heavy lifting required.
- **Environment Flexibility**
Configuration is managed via .env files, making it easy to switch between development and production environments.

Getting Started

Here's how to get SynapseStocks up and running in just a few steps:

1. Clone the Repository
2. `git clone https://github.com/Cognizant-SPE/vibe-coding-synapsestocks.git`
3. `cd vibe-coding-synapsestocks`
4. Set Up the Backend
5. `cd server`
6. `npm install`
7. `cp .env.sample .env` # or create your own .env
8. `npm run dev`
9. Set Up the Frontend
10. `cd ../client`
11. `npm install`
12. `cp .env.sample .env` # or create your own .env
13. `npm start`

PROJECT IMPACT & IMPLEMENTATION